

Game Design Documentation

Alien Outbreak: The Dark House

Game Overview

Alien Outbreak: The Dark House is a 3D OpenGL survival, exploration, and rescue game that takes place inside a large house consisting of multiple rooms. The house has been invaded by hostile aliens and infected humans. The entire environment is dark, creating a limited-visibility experience where the player must use a flashlight to explore the house and identify threats.

The player has two main responsibilities throughout the game. **Red aliens must be eliminated**, while **green aliens must not be killed because they are actually infected humans**. Instead, the player must use a chemical weapon to cure them and transform them back into humans.

However, curing a human does not immediately mean that the human has been rescued. Once a green alien is cured, all remaining aliens temporarily shift their attention toward that human and attempt to reach them. The player must quickly reach and touch the human before any alien does. If the player reaches the human first, the human is saved. If an alien reaches the human first, the human dies.

The player's objective is to successfully rescue **10 humans**. After rescuing all 10 humans, the player can proceed to the final room of the house, locate the hole through which the alien invasion originated, and seal it. Sealing the hole completes the game and results in victory.

To keep the project aligned with the requirement that the final project should contain approximately three times the number of major features compared with Assignment 3, the entire game is organized into **9 major features**.

Feature 1: 3D Multi-Room House Exploration System

The game will take place inside a fully constructed 3D house rather than a single open arena. The house will contain multiple separate rooms connected through entrances or doorways. The exact number, design, and layout of the rooms can be finalized later.

Each room will have its own:

- Floor
- Walls
- Boundaries
- Entrances or doors
- Fixed number of red aliens
- Fixed number of green aliens

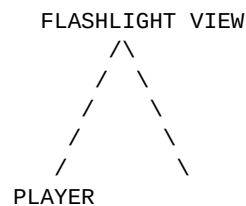
The player will move through the house and explore the rooms one by one. Unlike the previous assignment, where the player was restricted to a simple square arena, this project will require the player to navigate through a structured environment. Collision with walls and boundaries will prevent the player from moving outside the playable area or passing through the house structure.

Each room will contain a predetermined number of aliens. The number of red and green aliens in each room will be decided later during the detailed level design. The final room will contain the **alien entry hole**. This hole will not continuously generate or spawn aliens; its purpose is purely to serve as the final objective of the game.

Feature 2: Darkness and Flashlight Visibility System

The entire house will be dark. The player will not be able to clearly see the environment, rooms, or aliens without using a flashlight or torch.

The flashlight will have a limited viewing range, direction, and visibility area. It will follow the direction in which the player is facing, and only objects and aliens located within the flashlight's effective viewing range will become visible to the player.



The flashlight is therefore not only a visual element but also an important gameplay mechanic. The player must actively search the dark environment to locate aliens and determine whether they are red or green. However, pointing the flashlight toward an alien also creates danger because an illuminated alien can enter a combat state.

- Without the flashlight, the player cannot easily see approaching aliens.
- With the flashlight, the player can identify and attack aliens.
- Illuminated aliens can react and become dangerous in combat.

Feature 3: Dual Perspective and Player Control System

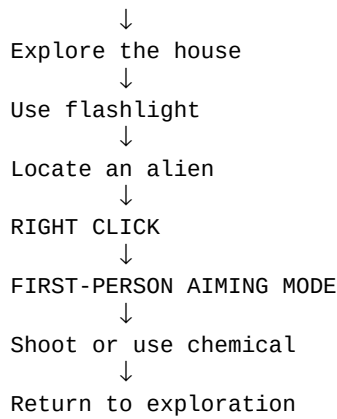
The game will contain two different perspectives.

Third-Person Perspective: The game will primarily be played in Third-Person POV. In this mode, the player character is visible, can move around the house, rotate and change direction, and use the flashlight to search for red and green aliens. This perspective is mainly used for exploration and movement.

First-Person Shooting Perspective: When the player right-clicks, the perspective will change from Third-Person POV to a First-Person shooting mode. This mode will function similarly to a sniper or aiming perspective.

- The player cannot move.
- The player can aim at visible targets.
- A mouse pointer or crosshair will be used for aiming.
- The player can shoot at aliens or use the chemical weapon.
- The player can switch back to Third-Person mode.

THIRD-PERSON MODE



The player cannot move while using First-Person aiming mode, creating danger because aliens can continue approaching while the player is stationary and attempting to aim.

Feature 4: Alien AI and Dynamic Movement System

The game will contain two main types of aliens: red aliens and green infected humans. Both types will roam throughout their assigned rooms.

Their movement will not simply involve moving directly toward the player in a straight line. Instead, the aliens will move in a circular or orbit-like motion while gradually decreasing their distance from their target.

This behavior will occur whether the flashlight is currently pointed at the alien or not. Therefore, avoiding the flashlight does not make the player safe; an alien hidden in darkness can still continue approaching the player.

Normal State: The alien roams in circular movement, slowly approaches the player, does not fire projectiles, and can instantly kill the player through physical contact.

Combat State: When illuminated by the flashlight, the alien becomes visible, the player can target it, and the alien can begin firing back while continuing its movement behavior.

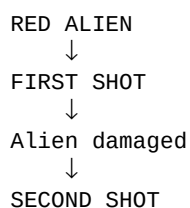
Human Target State: When a green alien is cured and transformed into a human, all remaining aliens temporarily stop targeting the player, stop firing, and begin approaching the newly created human.

Feature 5: Red Alien Combat and Gun System

Red aliens are fully hostile aliens and must be eliminated. During their normal roaming state, they slowly approach the player. If a red alien physically touches the player, the player immediately dies and the game ends.

When a red alien is illuminated by the flashlight, it enters combat mode. In combat mode, the player can shoot the red alien, while the red alien can fire projectiles toward the player.

Initially, every red alien will require **two successful shots** from the normal gun to be eliminated.



↓
Alien eliminated

The game will also include a gun upgrade or gun switching system.

Normal Gun: Standard damage; a red alien requires 2 successful shots to die.

Upgraded Gun: Higher damage; a red alien can be killed with only 1 successful shot.

Red aliens will also have a projectile system. Their projectiles can damage the player, while direct physical contact with any alien will result in immediate death.

Feature 6: Green Alien and Chemical Curing System

Green aliens are not actually normal aliens. They are **infected humans**. Although they are dangerous and can kill the player if they make physical contact, they must not be killed using the normal gun.

Green aliens will have the same general movement system: circular or orbit-like roaming, gradual movement toward the player, continued movement even when hidden in darkness, and immediate player death upon physical contact.

Instead of shooting a green alien with the normal gun, the player must use a **chemical weapon**. The chemical will function as a projectile similar to a bullet.

```
GREEN  INFECTED  HUMAN
      ↓
CHEMICAL  PROJECTILE
      ↓
HUMAN  CURED
      ↓
HUMAN
```

After a successful chemical collision, the green alien is removed or changes state, a human appears at the corresponding location, and a special rescue event begins. The human is not automatically counted as saved; the player must physically reach them.

Feature 7: Dynamic Human Rescue and Alien Target-Shifting System

This is one of the most important and unique mechanics of the game. When a green alien is successfully cured, the newly created human immediately attracts the attention of all remaining aliens.

```
Before:
ALL ALIENS
  ↓
PLAYER

After:
ALIENS → HUMAN ← PLAYER
```

During this rescue phase, aliens stop targeting the player, stop firing projectiles, temporarily stop their normal combat behavior, and slowly move toward the human. The player must race toward the human.

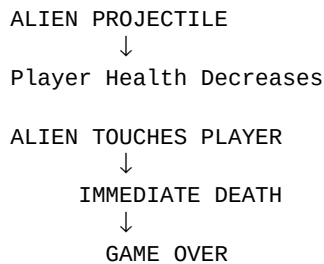
If the player reaches the human first: The human is successfully rescued, the rescue counter increases by one, the active rescue event ends, and all remaining aliens return to their original behavior and target the player again.

If an alien reaches the human first: The human dies, the rescue opportunity is lost, and the game returns to its normal state. The player must continue searching for other infected humans.

The player must repeat this process until **10 humans have been successfully rescued**.

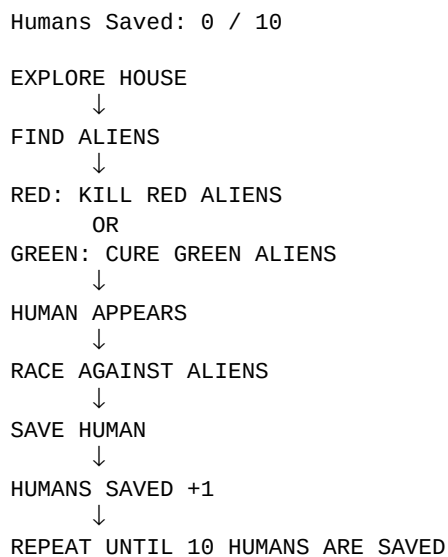
Feature 8: Player Survival, Health, and Game Progression System

The game will contain multiple survival and progression conditions. The player will have health-related mechanics for damage received from alien projectiles or other future hazards. However, physical contact with an alien will always be fatal.



The HUD will communicate important information such as player health, humans saved, current objective, current weapon, and gun status.

One of the primary progression requirements is:



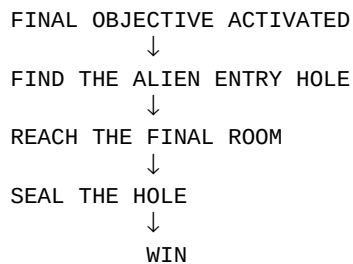
If the player dies, the game will enter a Game Over state and a restart option will be available.

Feature 9: Final Objective, Alien Hole, and Victory System

After the player successfully rescues 10 humans, the final objective becomes available. The player will then proceed toward the final room of the house, which contains the **alien entry hole**.

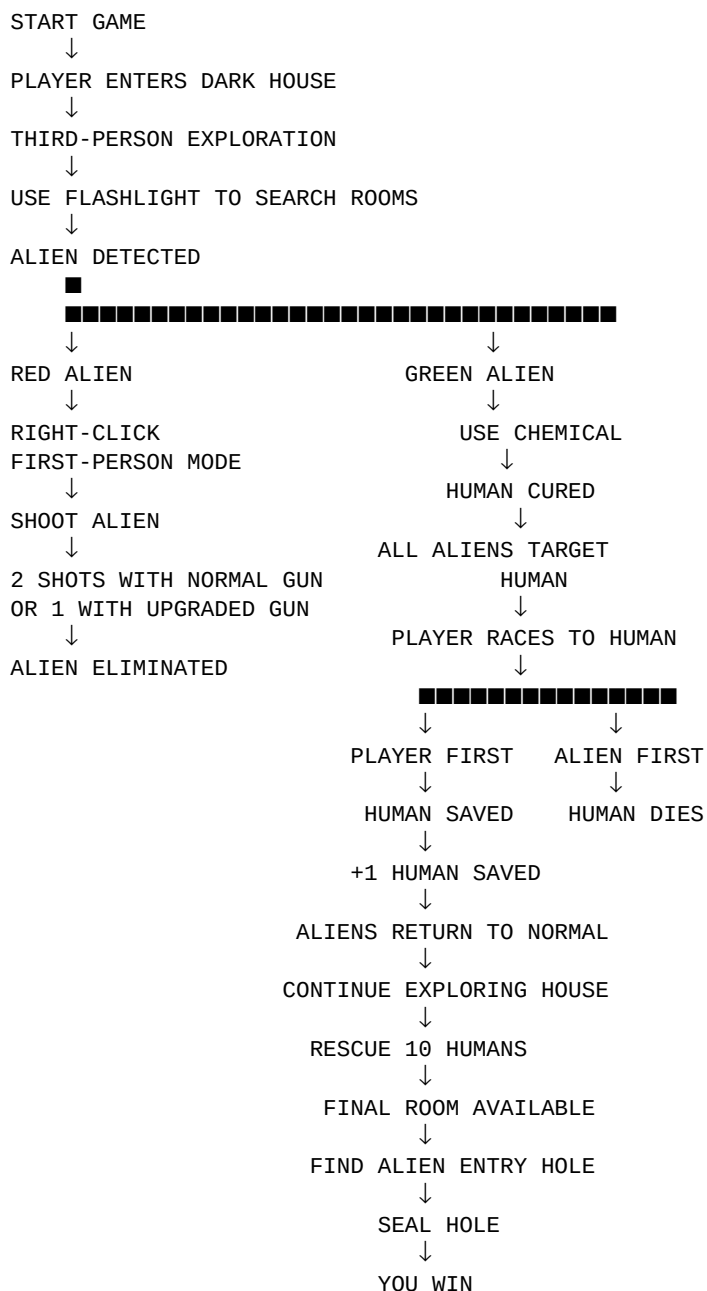
The hole represents the source or entry point of the alien invasion. However, it will not spawn additional aliens. The number of aliens in the game will be determined by the fixed alien distribution across the rooms.





Once the hole is successfully sealed, the game ends with a victory state. A victory screen can display information such as the number of humans saved, the number of red aliens eliminated, the final weapon used, and overall mission completion.

Complete Game Flow



Final Major Feature Summary

1. 3D Multi-Room House Exploration System
2. Darkness and Flashlight Visibility System
3. Dual Perspective and Player Control System
4. Alien AI and Dynamic Movement System
5. Red Alien Combat and Gun System
6. Green Alien and Chemical Curing System
7. Dynamic Human Rescue and Alien Target-Shifting System
8. Player Survival, Health, and Game Progression System
9. Final Objective, Alien Hole, and Victory System

This structure presents the final project as exactly **9 major features**. Since Assignment 3 was considered to have 3 major features, the project satisfies the intended 3× feature requirement through 9 clearly defined major systems.